Replication of Experiment 2 by Simon Farrell
(2008, Journal of Experimental Psychology: Learning, Memory, and Cognition)

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Introduction

When recalling sequences of items from short-term memory, people are able to recall the order of the presented items as well as the timing with which they were presented. It is not known whether memory for order and memory for timing rely on the same underlying system, or even the degree to which these two aspects of short-term memory affect one another. The original study by Simon Farrell (2008, JEP:LMC) investigated the claim that memory for order and timing rely on the same mechanisms. Three main findings were presented with roughly equal emphasis. First, instructing participants to group the items induced a temporal gap between the groups in both the order task and the timing task (thus impairing timing task performance); second, the input timing was reflected in the output timing in both the order task (when it was irrelevant) and the timing task; and third, temporally isolated items were recalled more accurately only when the recall task was precued, and not when postcued. The result we have selected as our target for replication is the first one, showing that asking participants to group stimuli alters the timing they report, even when subjects are asked to report timing faithfully. This result strongly suggests a dependence between the systems for order and timing: people's representation of order affects their representation of timing even when this crossover is detrimental to task performance.

Methods

Power Analysis

The grouping effect on output timing was reported in Experiment 2 using a repeated-measures, 2 (grouped vs. ungrouped) x 6 (serial position) ANOVA to reveal a significant interaction. Additionally, a more specific test based on deviations from a polynomial fit was conducted to verify that the interaction was actually due to a discontinuously elevated latency in the 4th reported item.

The ANOVA reported in the original paper showed that the interaction between grouping instruction and serial position was significant, $F(4.09, 159.50) = 4.03$, $p = .004$, $\eta_p^2 = 0.09$. As calculated using the ANOVA: Repeated Measures, Within Factors tool in G*Power 3.1.3, this value of η_p^2 corresponds to an effect size $f = 0.314$. The nonsphericity correction applied to the original analysis was determined from the reported degrees of freedom to be $\varepsilon = 0.818$. The correlation among repeated measures was not known or reported, so we investigated all possible correlation values between -0.5 and 0.5. Within this window, the largest suggested sample sizes were at a correlation of -0.5. The samples

required to achieve 80%, 90%, and 95% power as calculated using G*Power with a -0.5 correlation value would be 40, 50, and 60 subjects respectively.

Further analysis tested whether or not increased response latency occurred specifically at the group boundary. The original study author recommended that we use this result as the basis of our power analysis, because ultimately it more strongly supports the specific hypothesis put forth by the author about grouping. This additional analysis investigates the deviation of the fourth latency from what would be predicted from the surrounding latencies. This deviation analysis is conducted as follows: first, a second-order polynomial is fit to the average of each of the second, third, fifth, and sixth latencies in the grouped and ungrouped conditions for each participant; next, the deviations of the fourth latencies from the predictions of each polynomial are calculated; and finally, a t-test is run over 40 data points, each representing one subject's average fourth latency deviation, to determine whether the fourth latencies are significantly elevated. The t-test reported in the original paper showed the deviation of the fourth latency to be significant, $t(39) = 3.77$, $p < 0.001$, $d = 0.60$. A G*Power analysis run on t-test showed that 80%, 90%, and 95% power would be achievable with 24, 32, and 39 subjects respectively.

Planned Sample

We intend to test 40 participants, as this is the number of participants needed to ensure a 95% power on the most relevant grouping deviation measure assuming that the correlation among repeated measures falls between -0.5 and 0.5, and still ensures 80% power on the ANOVA measure. Participants will be solicited from the departmental mailing list of potential test subjects. This population includes members of the MIT community and other residents of the Greater Boston area. As in the original study, all participants will be native or fluent English speakers, and each participant will complete trials in all cells of the 2 (task type: order vs. timing) x 2 (instructions: ungrouped vs. grouped) x 6 (serial position: 1– 6) design.

Materials

The original study describes the materials for Experiments 1 and 2 as follows:

> "Lists contained six digits that were randomly sampled without replacement from the 1–9 set. Sixty lists were constructed for each recall task and were subject to two constraints: (a) Following Henson (1996), lists could not contain ascending or descending pairs of integers (e.g., "3 4" or "7 6"), and (b) an item could not appear in the same serial position on consecutive lists."

Furthermore, specific to Experiment 2,

> "The stimuli themselves were recorded from a male speaker with a neutral English accent and were digitized for storage on hard disk. We pitch-flattened the digits to a fundamental frequency of 113 Hz using the phonetics software Praat (Boersma & Weenink, 2006)—to avoid

any grouping effects arising from pitch variations (see Nicholls & Jones, 2002)—and we normalized the digits to a constant length of 500 ms. The digits were additionally padded to give a regular sounding sequence for different permutations, as it is known (and was observed in pilot testing) that the subjective experience of the center of an acoustic stimulus does not exactly relate to the onset of the stimulus (e.g., 137 Scott, 1998). The average duration of this padding was 85.9 ms (SD б 50). […] Raw pause durations were generated by pseudorandomly sampling from a uniform distribution, and the durations on each list were then scaled such that durations summed to a constant 3,000 ms."

The original study was performed on PC with a 15-in. [38.1 cm] monitor and using the Psychophysics Toolbox for MATLAB (Brainard, 1997; Pelli, 1997). We intend to use the original MATLAB code, including the original audio stimuli. Therefore we will adhere to these methods almost exactly, with the sole exception that our monitor size may be different.

Procedure

The original study describes the shared procedure for Experiments 1 and 2 as follows:

"Participants were tested individually in a laboratory. Each trial began with a fixation point (a cross) being presented in the center of the screen for 1,000 ms. This was followed by a blankscreen with a duration of 500 ms, which was then followed by presentation of the memory list. […] Following presentation of the list, there was another blank screen of 500 ms, followed by presentation of one of the two cues "NUMBERS" and "RHYTHM." If "NUMBERS" appeared on the screen, the participant's task was to recall the digits from the list in the order they were presented by typing them on the numeric keypad of the keyboard (i.e., standard serial recall). If the "RHYTHM" cue appeared, the participants were instructed to ignore the identity of the digits, and instead to tap the "5" key on the numeric keypad in time with the remembered timing of the input list; that is, they should tap the key at the remembered onset of each item. In effect, the timing task required participants to reproduce the remembered durations of the items on the memory list in order. Demonstration trials were given to show exactly what was required of the participants in the timing task. The task cue remained on the screen throughout the recall period. When the participant had pressed six keys on the keypad, the cue was replaced by a blank screen for 1,000 ms before moving on to the next trial. For the first half of the experiment […] the participants carried out the experiment in line with the procedure as detailed. At the halfway point a message appeared on the screen with additional instructions. These instructions were to "think about the six numbers on each list as being in two groups of 3"; participants were given an example of such a grouping ("For example, if you got the list of numbers '1 7 4 8 3 2,' you would remember '1 7 4' as one group, and '8 3 2' as another group"). The experimental task was otherwise unchanged. The grouping instruction was presented in the second half of the experiment for all participants. Although this leads to an order confound, this constant ordering of conditions was chosen because the grouping strategy was otherwise expected to continue after people had been presented with grouped

> lists and thus contaminate the ungrouped condition (Farrell & Lewandowsky, 2004; Henson,1999). Four practice trials were presented before the experimental session began (two order trials and two timing trials) to familiarize participants with the experiment. A self-paced break was presented after every 10 experimental trials; participants were presented with a message indicating that they should press the space bar to move to the next set of trials. The entire procedure lasted for around 40 min."

The description of how the memory list was presented in Experiment 2 is as follows:

> "List stimuli were played over headphones to participants at a comfortable volume determined prior to the experiment. Forty lists were presented under the standard instructions and grouping instructions, 20 each for the order task and timing task, respectively."

This procedure will be followed precisely.

Analysis Plan

The dependent variable in this study is the response latency for each of the six items reported by subjects in each timing recall trial. In each trial, the time interval before each of the subjects' responses will be measured. Subjects will perform both order recall trials and timing recall trials so that they will not know in advance what they will be asked to report, but only the latencies from the timing trials will be used in the analysis. Subjects will first perform trials in the ungrouped condition; after being instructed to think of the items in two groups of three, they will then perform the remaining trials in the grouped condition.

As reported in the methods section, the central claim about the effect of grouping on timing is testable in two key ways. The first, less specific test is to run an ANOVA to determine if grouping had any significant effect on subjects' latencies. To look for an effect of any sort, a repeated-measures, 2 (grouped vs. ungrouped) x 6 (serial position) ANOVA will be performed. The presence or absence of an interaction will be the main statistic of interest: we expect to find a significant interaction between grouping condition and serial position. The second, more specific test is to look specifically for the hypothesized elevation of the latency of the fourth item (that is to say, to look for a gap between the first group and second group of numbers in subject's responses). To test for this specific effect, a test of deviation from polynomial fit will be performed.

Differences from Original Study

Our subjects will not all be undergraduates, and will be compensated monetarily rather than with course credit; we presume these differences will not alter the effect. Also, the original study did not provide information about subjects' age, gender, or other biographical details; we presume these differences are also irrelevant. We expect minimal differences in setting and procedure. As in the original study, subjects will be tested individually in a quiet laboratory setting, and we will use all the

original code and stimuli. We may use a different monitor size.

Methods Addendum

Actual Sample

Forty-five (45) participants were tested. Four subjects' results were dropped due to software crashes which prevented the subjects from completing the entire study. One subject's results were dropped because after reading the grouping instructions the subject reported that he was familiar with the study and knew the hypothesis being tested. We were prepared to drop data from subjects with a very poor accuracy rate on the order task (low enough to indicate they were not cooperating with the instructions) but no subjects had to be dropped for poor accuracy. The final sample of forty (40) subjects consisted of twenty-five (25) men and fifteen (15) women between the ages of 18 and 61, with an average age of 33.

Differences from pre-data collection methods plan

Our deviations from our methods plan were only in hardware used. We used a different monitor size than in the original study: rather than a 15" external monitor as was used in the original experiment, we presented the stimuli on the built-in screen of a 12.5" laptop. In addition, the laptop used to run the experiment did not have a separate numeric keypad, so participants were instructed to use the number row at the top of the keyboard.

Results

Data preparation

Data were gathered into a CSV file and analyzed using R.

Confirmatory analysis

In a 2 (instruction: grouped vs. ungrouped) x 6 (serial position) ANOVA of the timing task, our analysis revealed a significant effect of serial position, $F(5, 195) = 78.90$, $p < 2 \times 10^{-16}$, $\eta_p^2 = 0.67$; no main effect of grouping, $F(1, 39) < 1.00$, $p = 0.45$, $\eta_p^2 = 0.014$; and a significant interaction, $F(5, 195) = 2.54$, $p < 0.03$, $\eta_p^2 = 0.061$. For comparison, the original study likewise reported a significant effect of serial position in the timing task, $F(1.63, 63.90) = 64.87$, $p < 0.001$, $\eta_p^2 = 0.63$; no main effect of grouping instruction, $F(1, 39) < 1.00$, $\eta_p^2 = 0.001$; and a significant interaction, $F(4.09, 159.60) = 4.03$, $p < 0.004$, $\eta_p^2 = 0.09$. Figure 1 illustrates this pattern of latencies on the timing task.

Paired samples *t*-tests comparing the deviation of the fourth latency from the predictions of a polynomial fit in the ungrouped and grouped conditions of the timing task revealed a significant difference between conditions, $t(39) = 2.08$, $p = 0.044$, $d = 0.37$ (95% CI [0.01, 0.65]), with positive

deviations in the grouping condition and negative deviations in the non-grouping condition. Compare this with the results of the original study, which reported that fourth latency deviations were larger (more positive) under grouping conditions in the timing task, $t(39) = 2.82$, $p = 0.008$, $d = 0.44$. Figure 3 illustrates the lower-than-predicted deviation on the third latency and the higher-than-predicted deviation on the fourth latency.

Exploratory analysis

Third latency analysis of the timing task. Visual analysis of the latency – serial position curve suggested a much larger discrepancy in the third latencies than in fourth latencies between the grouping and non-grouping conditions. Paired samples *t*-tests comparing the deviation of the *third* latency from the predictions of a polynomial fit in the ungrouped and grouped conditions of the timing task revealed a significant difference between conditions, $t(39) = -3.96$, $p < 0.0004$, $d = -0.65$, with negative deviations in both conditions but larger (more negative) deviations in the grouping condition.

Order condition analysis of latency. Although our primary replication target was the timing task, we collected data on the order task as well which can just as well be analyzed for the same effects. Figure 2 illustrates the pattern of latencies on the order task. A 2 (instruction: grouped vs. ungrouped) x 6 (serial position) ANOVA of the order task revealed a significant effect of serial position, $F(5, 195) = 122.9$, $p < 2 \times 10^{-16}$, $\eta_p^2 = 0.76$; no main effect of grouping, $F(1, 39) < 0.10$, $p = 0.81$, $\eta_p^2 = 0.0015$; and no interaction, $F(5, 195) = 1.20$, $p = 0.31$, $\eta_p^2 = 0.03$. For comparison, the original study reported a main effect of serial position in the order task, $F(1.95, 76.07) = 168.76$, $p < 0.001$, $\eta_p^2 = 0.81$; no effect of grouping instruction, $F(1, 39) < 1.00$, $p = 0.68$ $\eta_p^2 < 0.01$; and a significant interaction, $F(3.69, 143.77) = 6.96$, $p < 0.001$, $\eta_p^2 = .15$.

Paired samples *t*-tests comparing the deviation of the *fourth* latency from the predictions of a polynomial fit in the ungrouped and grouped conditions of the order task revealed a significant difference between conditions, $t(39) = 3.03$, $p = 0.004$, $d = 0.61$, with positive deviations in both conditions deviations and larger (more positive) deviations in the grouping condition. Compare this with the results of the original study, which reported that fourth latency deviations were larger (more positive) under grouping conditions in the order task, $t(39) = 5.52$, $p < .001$, $d = 0.87$.

Paired samples *t*-tests comparing the deviation of the *third* latency from the predictions of a polynomial fit in the ungrouped and grouped conditions of the order task revealed a significant difference between conditions, $t(39) = -3.73$, $p = 0.0006$, $d = -0.73$, with negative deviations in both conditions but larger (more negative) deviations in the grouping condition. Results from the original study are not available. Figure 4 illustrates the lower-than-predicted deviation on the third latency and the higher-than-predicted deviation on the fourth latency.

Analysis of accuracy

We investigated the accuracy of participants' responses in the order task. For this analysis, we were forced to drop the first five subjects because the identity of their keypresses was not recorded due to a software error. Figure 5 illustrates participants' performance.

Our participants' performance was comparable to the performance of participants in the original study. On average, participants had a mean item-level accuracy of 0.896 in both the grouped and ungrouped conditions. For comparison, the original study reported an ungrouped mean of 0.88 and a grouped mean of 0.92.

A 2 (instruction: grouped vs. ungrouped) x 6 (serial position) ANOVA of the accuracy responses in the order task revealed a significant effect of serial position, $F(5, 170) = 27.34$, $p < 2 \times 10^{-16}$, $\eta_p^2 = 0.45$; no main effect of grouping, $F(1, 34) < 0.01$, $p = 0.965$, $\eta_p^2 = 0.0001$; and no interaction, $F(5, 170) = 0.73$, $p = 0.602$, $\eta_p^2 = 0.021$. For comparison, the original study reported a main effect of serial position, $F(2.37, 92.48) = 28.32$, $p < 0.001$, $\eta_p^2 = 0.42$; a significant effect of grouping instruction, $F(1, 39) = 9.11$, $p = 0.004$ $\eta_p^2 = 0.19$; and no interaction, $F(3.10, 120.76) = 1.95$, $p = 0.124$, $\eta_p^2 = .05$.

Discussion

Summary of Replication Attempt

We were successful in replicating the finding that asking participants to group stimuli alters the timing they report, even when subjects are asked to report timing faithfully. The confirmatory analysis revealed a significant interaction between grouping instruction and serial position, as well as a significant difference in the deviations of the fourth latency from the predictions of a polynomial fit in the ungrouped and grouped conditions. Therefore, this study has replicated the original result.

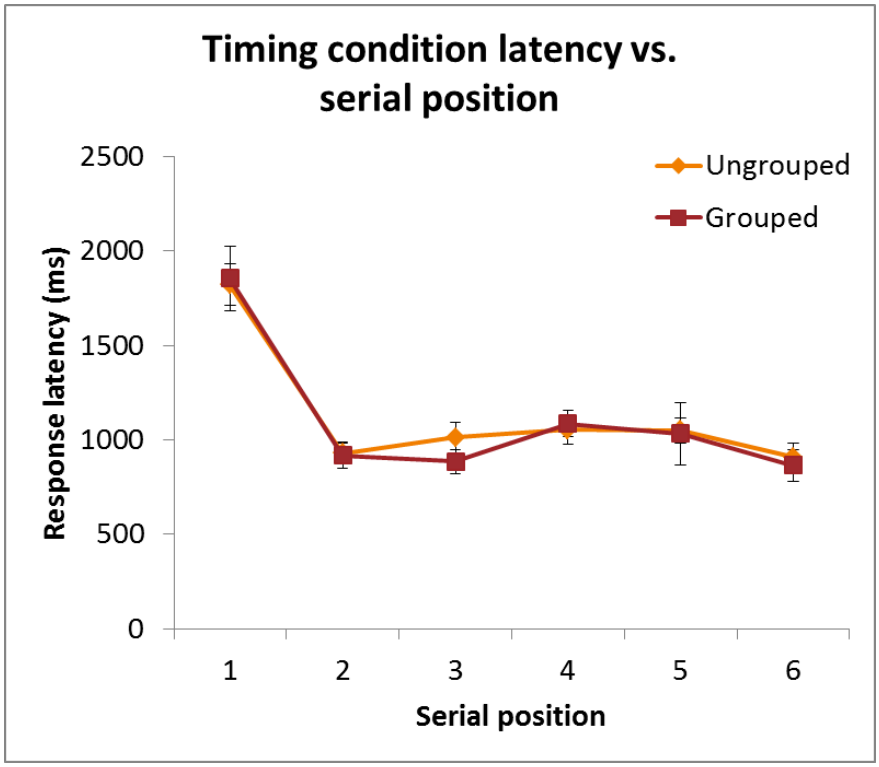


Figure 1: Latencies in milliseconds are plotted for serial positions 1 through 6 on the timing task (with standard error).

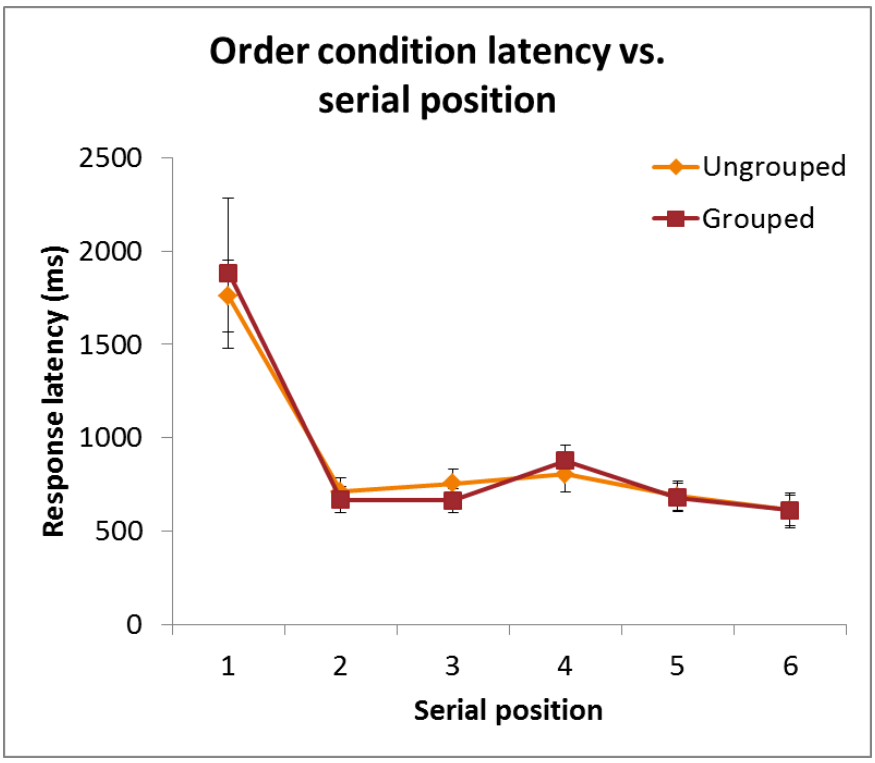


Figure 2: Latencies in milliseconds are plotted for serial positions 1 through 6 on the order task (with standard error).

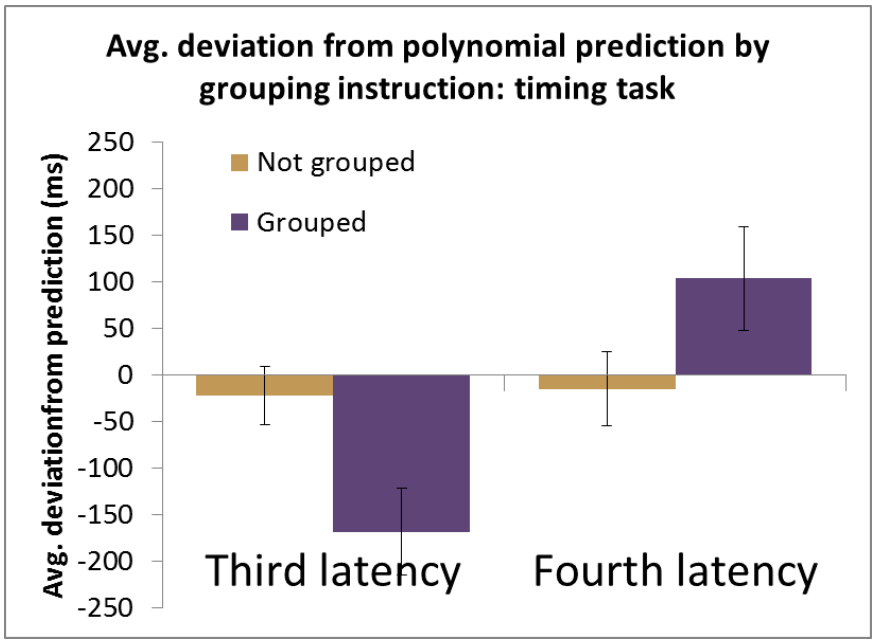


Figure 3: Average deviations of the third and fourth latencies from a polynomial fit of the timing task are plotted (with standard error). The key result we sought to replicate was that the grouping manipulation affects the timing of participants' responses, even when that effect is deleterious to performance of the explicit task.

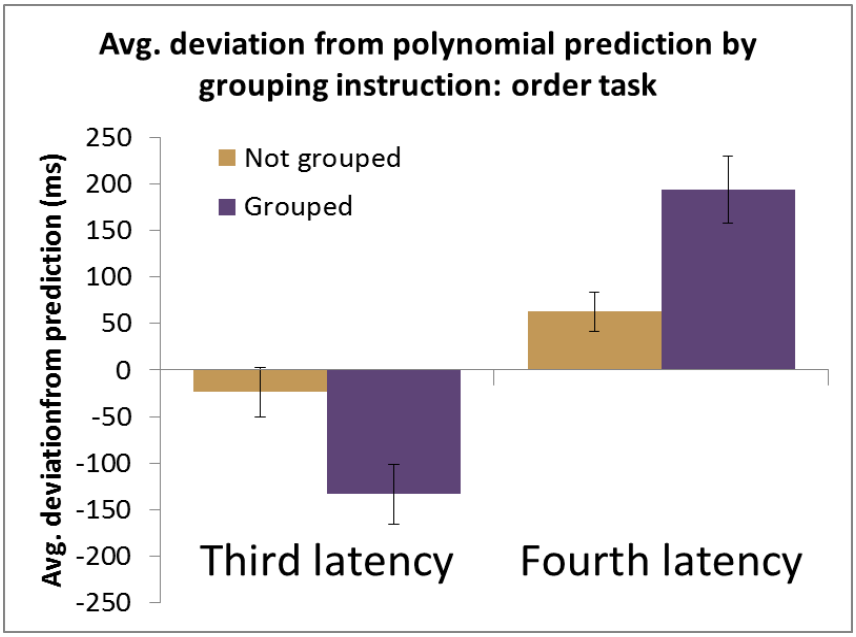


Figure 4: Average deviations of the third and fourth latencies from a polynomial fit of the order task are plotted (with standard error). In exploratory analysis, we sought to replicate the result that grouping manipulation affects the timing of participants' responses when that effect is unrelated to performance of the explicit task.

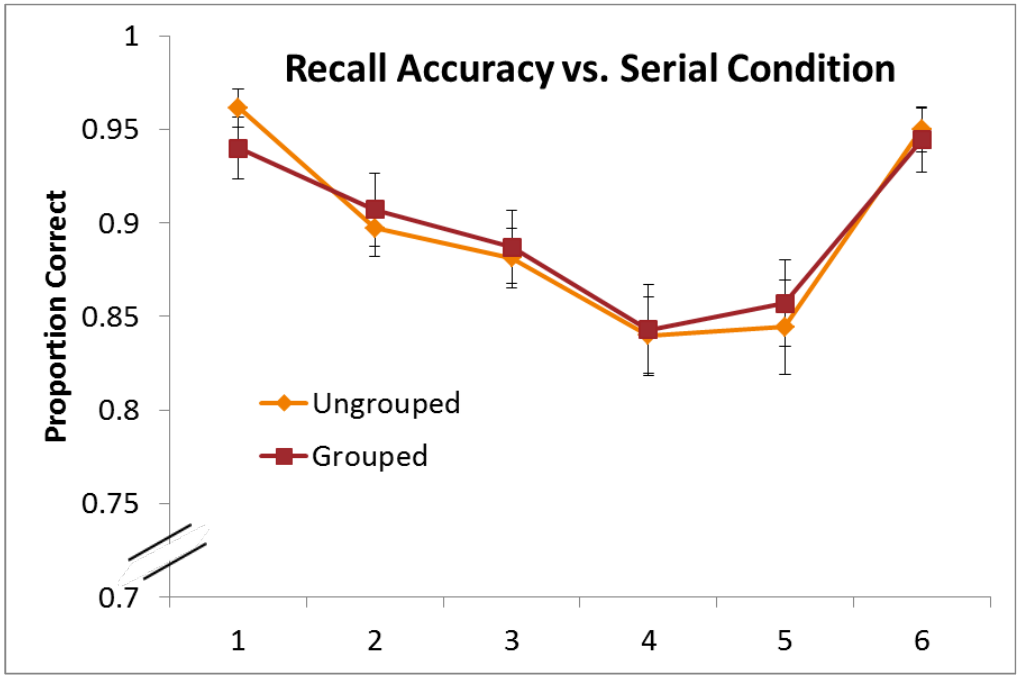


Figure 5: Average proportion of correct responses for ungrouped and grouped lists (with standard error).